Micro-Aeration for Improving Anaerobic Treatment and Biogas Production from Organic Pollutants

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Table S1 Summary of studies that applied micro-aeration on anaerobic digestion for improving the hydrolysis, including key parameters such as substrate, reactor configuration, air/oxygen dosage, and observed effects

Substrate	Reactor type	Air/O_2	Pre-treatment (Pt) Direct (D)	Air/O_2 Dose	Effect	Reference
Primary sludge	Batch bottles	Air	Pt - Semi-continuous (time-based)	0.21 L L^{-1} d^{-1}	Increased hydrolysis (50-60%); Reduced CH_4 yield	(Johansen and Bakke, 2006)
Primary sludge	Batch Flasks	Air	D - Continuous	0.5 L L^{-1} d^{-1}	Increased hydrolysis	(Diak et al. 2013)
40% primary sludge and 60% WAS	CSTR	Air	D - Pulses	12.5 and 25 mL L^{-1} d^{-1}	Increased biogas production (16±7%)	(Zhou et al. 2021)
WAS	CSTR	Air	D - Continuous	24-121 L L^{-1} d^{-1}	Increased solubilization	(Hasegawa et al. 2000)
WAS	CSTR	Air	D - Continuous	0.03 L L^{-1} d^{-1}	Decreased VFA; Decreased foaming; Better dewaterability	(Jenicek et al. 2014)
Mixed WAS and fly ashes	Batch bottles	Air	Pt - Continuous (48h)	0.35 vvm for 48 h	Increased CH_4 production (70%)	(Montalvo et al. 2018)
Wastewater primary sludge and WAS	CSTR	Air	D - Continuous (4h)	0-6 Air volume g^{-1} TS min^{-1}	Increased VFA; Increased biogas production (16%)	(Ruan et al. 2019)
Mixed sewage sludge	Reactor	Air	Pt - Continuous (48h)	0.3 vvm	Increase hydrolysis; Increased biogas	(Montalvo et al. 2016)
Sewage sludge	Batch bottles	Air	Pt - Semi-continuous	0.1, 0.2, 0.3 and 0.5 vvm (20h)	VSS reduction (31%); Increased microbial activities (597 %); Increased CH_4 production (221 %)	(Barati rashvanlou et al. 2020)
Activated sludge	AnMBR	Air	D - Continuous (ORP based)		Increased CH_4 production; Improved sludge filterability	(Wei et al. 2024)

Table S1 (cont.)

Substrate	Reactor type	Air/O_2	Pre-treatment (Pt) Direct (D)	Air/O_2 Dose	Effect	Reference
Synthetic wastewater with sucrose	Bench scale reactors	Air	D - Semi-continuous (time-based)	0.07 and 0.7 L L^{-1} d^{-1}	Increased COD removal; More stable pH	(Zitomer and Shrout 1998)
Synthetic wastewater (starch, peptone, yeast extract, phosphate salts)	Batch reactor	Air	D - Intermittent (daily pulses)	0, 2.52, 5.04 and 10.07 (as % of feed COD)	Decrease of CH_4 /biogas; Decrease of VFA accumulation; More stable operation	(Botheju et al. 2010)
Wastewater treatment plant	Pilot scale bioreactors	Air	Pt - Continuous	DO level of 3.0-4.0 mg L^{-1}	Decrease sludge yield	(Niu et al. 2016)
Industrial wastewater and domestic sewage	Full scale anaerobic–anoxic–aerobic with a microaerobic tank	Air	D - Continuous	0.5-1.0 mg L^{-1}	Low sludge yield (0.074 g SS g^{-1} COD); High NH_4-N, TN and COD removal	(Jiang et al. 2018)
Lipid rich slaughterhouse wastewater	Bottles	Air	Pt - Continuous (through feeding)	0.05 vvm and 0.15 vvm	Reduced TSS; Increased digestibility; Increased CH_4 production yields	(Zouari and Al Jabiri, 2015)
Brown water and food waste	Batch reactors	Air	Pt - Continuous	0.04 L L^{-1} d^{-1}	Increased hydrolysis and acidogenesis; Increased CH_4 (21%)	(Lim and Wang, 2013)
Brown water and food waste	CSTR	O_2	D - Single dosing	0.005 and 0.007 L L^{-1} d^{-1}	Increased hydrolysis; Increased acidogenesis	(Lim et al. 2014)
Food waste	Semi-continuous digester	Air	D- Semi-continuous	5 mL air per feeding (corresponding to an increase of 30–50 mV in ORP)	Enhanced CH_4 yield; Improved system stability; Increased microbial activity; Reduced dissolved organic matter	(Ding et al. 2024)

Table S1 (cont.)

Substrate	Reactor type	Air/O_2	Pre-treatment (Pt) Direct (D)	Air/O_2 Dose	Effect	Reference
Kitchen waste	SBR	O_2	Pt - Single dosing	1.11 mL g^{-1} VS d^{-1})	Increased biogas production	(Zhen et al. 2020)
Grass silage	Leach bed reactor	Air	D - Semi-continuous (time-based, 24h)	0.5 and 5 L L^{-1} d^{-1}	Increased VFA production (4x); Extensive aeration: VFA reduction	(Jagadabhi et al. 2010)
Napier grass	Batch bottles	O_2	D - Single dosing	15 mL gVS^{-1} = 0.09 and 1.9 L L^{-1} d^{-1}	Increased VFA production	(Sawatdeenarunat et al. 2017)
Napier grass	CSTR	O_2	C - Intermittent (every 24h) (ORP-based)	0.2 L L^{-1} d^{-1}	No VFA accumulation; Increased biogas production	(Nguyen, et al., 2019)
Corn straw	Batch bottles	O_2	Pt - Single dosing	5 mL g^{-1} VS^{-1} substrate	Increased hydrolysis; Increased CH_4 yield (16 %); Reduced lag phase	(Fu et al. 2015)
Corn straw	Batch bottles	O_2	Pt - Single dose	5 mL O2 g^{-1} VS	Increased CH_4 yield (271 mL g^{-1} VS); Increased glucose and VFA production	(Xu et al., 2018)
Corn straw	Batch bottles	O_2	Pt - Nanobubbles		Increased lignocellulosic hydrolysis (20-38%); Increased CH_4 yield (10-20%)	(Wang, et al., 2020) b
Corn derived wastewater (nejayote) + GAC	2 stage system APBR UASB	Air	D - Intermittent (time-based)	0.7 L min^{-1}	Increased VFA production (62%); Increased CH_4 yield (55%)	(Valero et al. 2020)
Dry corn stover	NM	Air	Pt -Continuous - Feeding open to air	0.1-1 mg L^{-1}	Increased hydrolysis; Increased AD rate	(Xu et al. 2021)
Wheat straw	Batch reactor	O_2	D - Intermittent (time-based)	5 mL O_2 g^{-1} VS	Increased hydrolysis; Increased Biogas	(Tsapekos et al. 2017)

Table S1 (cont.)

Substrate	Reactor type	Air/O_2	Pre-treatment (Pt) Direct (D)	Air/O_2 Dose	Effect	Reference
Wheat straw	Batch reactor	O_2	D - Intermittent (time-based)	5 mL O_2 g^{-1} VS	Increased hydrolysis; Increased Biogas	(Tsapekos et al. 2017)
Municipal solid waste	Pilot scale Leach bed reactor	Air	D - Semi-continuous (time-based)	0.39 L L^{-1} d^{-1}	Increased CH_4 production Increased hydrolysis and acidification	(Nguyen et al. 2007)
Municipal solid waste	Landfill simulation reactors		D - Intermittent	0.025 L $h^{-1}g^{-1}$, 0.05 L h^{-1} kg^{-1}, and 0.1 L h^{-1} kg^{-1}	Increased degradation of VFA; NH_4^+-N and organic matter reduction	(Liu et al. 2024)
Synthetic food waste	Leach bed rector	Air	D - Semi-continuous (time-based)	2.1, 4.4 and 6.5 L L^{-1} d^{-1}	Increased carbohydrate hydrolysis (21-27%); Increased protein hydrolysis (38-64%); Increased CH_4 yield	(Xu et al., 2014)
Cellulose	Batch bottles	Air	D - Nanobubbles		Increased VFA production; Increased CH_4 yield (10-18%)	(Wang, et al., 2020)
Cellulose	Batch bottles	Air O_2	Pt - Nanobubbles	Air- NBW (193 NmL g^{-1} VSreduced) O_2-NBW (233 NmL g^{-1} VSreduced)	Reduced cellulose crystallinity; Increased cellulase production Increased CH_4 yield (8-30%);	(Wang et al. 2020c)
Paper waste	Batch bottles	O_2	Pt - Single dosing	0, 5, 15, and 30 mL g^{-1} VS	Increased hydrolysis and acetogenesis; Increased CH4	(Song et al. 2021)
Cow manure	Batch bottles	Air	D - Pulses	37.5 mL L^{-1} min^{-1}	Increased hydrolysis	(Li et al. 2023)
Fresh 'as excreted' buffalo manure	Batch bottles	Air	Pt - Single dosing	7.3 mL g^{-1} VS (12h)	Increased lignin degradation	(Zeb et al. 2022)
1) Swine manure 2) corn silage + swine manure	Batch	Air	D - Intermittent	0.2 L L^{-1} d^{-1}	Higher VFA; lower lag phase	(Cao et al. 2022)

COD, Chemical oxygen demand. CSTR, Continuous stirred tank reactor. GAC, Granulated activated carbon. NM, Not mentioned. WAS, waste activated sludge. AnMBR, Anaerobic membrane bioreactor. APBR, Anaerobic packed bed reactors. NBW, Nanobubble water. SBR, Sequencing batch reactor. UASB, up-flow anaerobic sludge blanket. VFA, volatile fatty acids. VS, volatile solids. VSS, volatile suspended solids. vvm, gas volume flow per unit of liquid volume per minute.

Table S2 Summary of studies that applied micro-aeration on anaerobic digestion for H_2S removal, including key parameters such as substrate, reactor configuration, air/oxygen dosage, and observed effects

Substrate	Reactor type	Air/O_2	Pre-treatment (Pt) Direct (D)	O_2 Dose	Effect	Reference
Primary sludge	CSTR	O_2	D - Semi-continuous (ORP-based)	0.14 mL s^{-1}	Removal of H2S (99 %) No effects in VS and COD removal	(Nghiem et al. 2014)
WAS	Pilot scale bioreactor	Air/ O_2	D - Continuous (sludge recirculation)	0.25 L L^{-1} d^{-1}	Removal of H2S (98%) COD removal, VS reduction ad CH4 yield not affected	(Díaz et al. 2010)
WAS	Pilot scale CSTR	O_2	D - Continuous	0.013-0.024 L L^{-1} d^{-1}	Removal of H2S (99% Without negative effects on CH4	(Fdz.-Polanco et al. 2009)
WAS	Pilot scale bioreactor	O_2	D - Continuous (Biogas, sludge recirculation or headspace)	0.01 L L^{-1} d^{-1}	Removal of H2S (98%)	(Díaz et al. 2011)
WAS	Pilot scale CSTR	O_2	D - Continuous	0.01 L L^{-1} d^{-1}	Overcome overloading; Removal of H2S (90%); Prevention of severe imbalance	(Ramos and Fdz-Polanco, 2013)
Sewage sludge	Pilot scale CSTR	O_2	D - Continuous (headspace, biogas recirculation, liquid)	0.002 %-0.20 % (v/v)	High removal of H2S No difference in the digestion Increase of community diversity	(Ramos et al. 2014)
Pulp mill wastewater	UASB	Air	D - Continuous (in the feeding immediately entering the reactor)	0.9-1.8 L L^{-1} d^{-1}	Removal of H2S (30%) Increased COD removal rate (50%)	(Zhou et al., 2007)
High solid wastewater	CSTR + SOU	Air	D - Continuous (ORP-based)	0.0782 L L^{-1} d^{-1}	Removal of H2S (98%)	(Duangmanee et al. 2007)
Synthetic wastewater with ethanol	Continuous-flow anaerobic reactors	Air	D - Continuous	0.2 mL min^{-1}	Reduced H2S (93%) Increased system stability	(Sousa et al. 2016)
Municipal wastewater	EGSB	Air	D - Sludge recirculation	DO of 0.2 mg L^{-1}	Removal H2S (70%)	(Chen et al. 2017)

Table S2 (cont.)

Substrate	Reactor type	Air/O_2	Pre-treatment (Pt) Direct (D)	O_2 Dose	Effect	Reference
Municipal wastewater	Full scale plants	Air	D - Sludge recirculation	0.056-1.2 m^3 h^{-1}	Removal H2S (74%-99%)	(Jeníček et al. 2017)
Sulfate-rich wastewater	Batch reactors	O_2	D - Continuous (Biogas recirculation, ORP based)	1-2 mL min^{-1}	Removal of H2S (>99) Increased COD removal (5 %); Increased CH4 rate production (56 %)	(Khanal and Huang, 2003)
Sulfate-rich wastewater	UAF	O_2	D - Continuous (Biogas recirculation, ORP based)	5-6 mL min^{-1}	Removal of H2S (>98.5 %) Increase CH4 rate production (46%) Airflow >25 mV CH4 negative impact	(Khanal et al. 2003)
Sulfate-rich wastewater	UAF	O_2	D - Semi-continuous (ORP-based)		Removal of H2S (<99%) Increased CH4 yield (46%)	(Khanal and Huang 2006)
Synthetic brewery wastewater	UASB	O_2	D - Continuous	0.08 L L^{-1} d^{-1}	Removal of H2S (73 %)	(Krayzelova et al. 2014)
Vinasse	Fluidized bed reactor	Air	D - Continuous	0.15-0.19 L L^{-1} d^{-1}	Removal of H2S (99%) Faster process	(van der Zee et al. 2007)
Cheese way	Pilot scale CSTR	Air	D - Continuous (through gas membrane)	0.8, 1.2, 2, 4, 40, 400 mL min^{-1}	Decrease of H2S (3000 ppm to <less than 100 ppm) COD not affected	(Pokorna-Krayzelova et al. 2018)
Cow manure	Full-scale digester	Air	D - Continuous (in the headspace)	1% (v/v) biogas production rate	Removal of H2S (68.2%)	(Kobayashi et al. 2012)

COD, Chemical oxygen demand. CSTR, Continuous stirred tank reactor. EGSB, expanded granular sludge bed. SOU, Sulfide oxidizing unit. UAF, Up-flow anaerobic filter. UASB, up-flow anaerobic sludge blanket. VS, volatile solids.

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